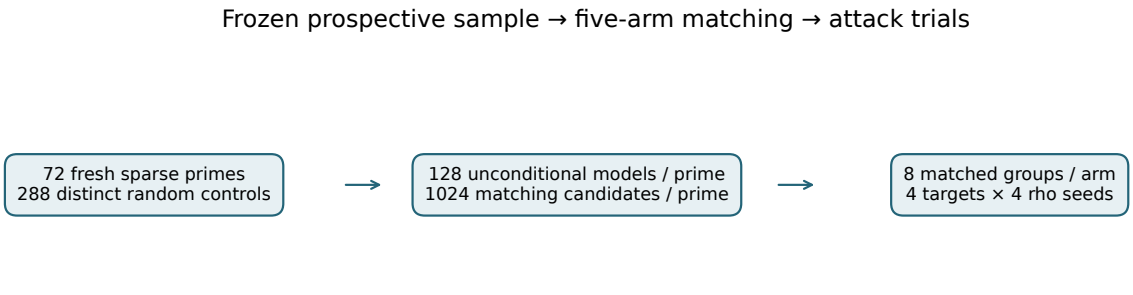
Wider Solinas prime comparison

2026-10-11 · EXP-ECDLP-5184f7 · RUN-ECDLP-07aad1

Native rho ratio 0.9952

Marginal **99.9% interval: 0.9725–1.0185**. Reserved three-family 99.9667% Bonferroni interval: 0.9706–1.0207. Frozen practical-similarity margin: 0.8333–1.2000. Decision: practically similar normalized rho cost in this sampled scope.

Prospective wider scope: 72 fresh sparse-prime blocks, 288 random controls, four field-bit strata (24/28/32/36), 368,640 nonsingular prepared models and 46,080 unconditional audit models. Baseline moduli are excluded and baseline outcomes are not pooled.

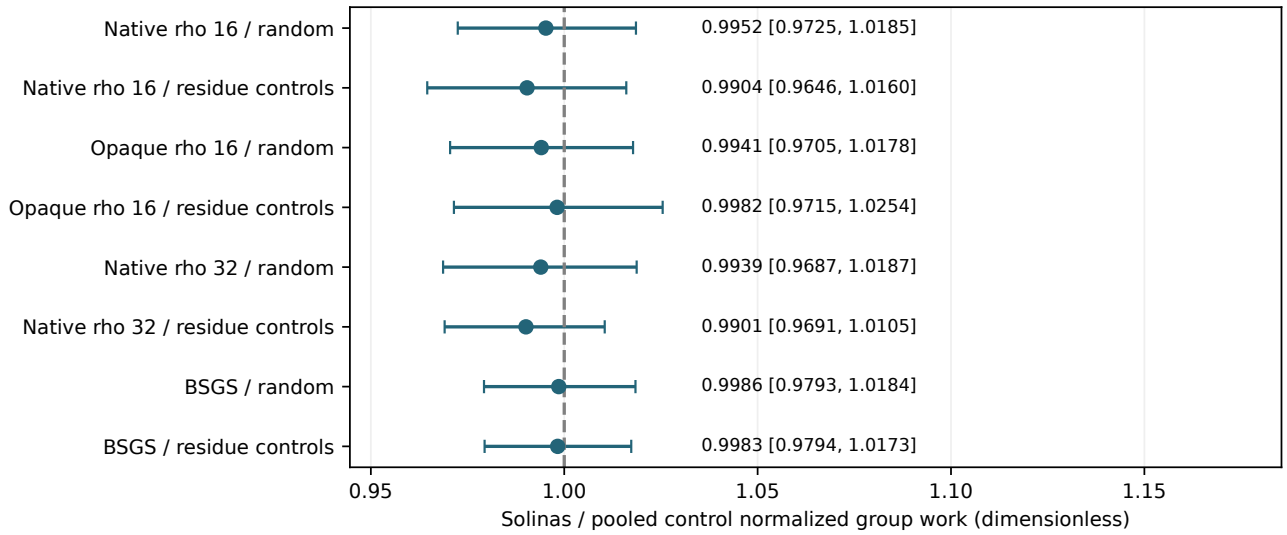


The sparse integer catalog is $p=2^b-2^a+s2^c+t$, $1 \le c \le a-1$, $a \le b-2$, $s,t=\pm1$. Each block has two size-matched random controls and two controls additionally matching p mod 12. Fixed pools give eight common S matches across all four control arms, within 10% subgroup order; largest ell has 16–28 bits. The unconditional audit does not exclude weak curves.

Whole-prime-block bootstrap: 500,000 replicates within bit-size strata. Targets and walk seeds are repetitions, not independent prime samples. Intervals have approximate nominal coverage. The three-family adjustment concerns primary tests only; other results use marginal intervals. This run does not identify or test the two other user-named prime cases.

Attack results and controls

72 fresh prime blocks · marginal 99.9% whole-block bootstrap intervals



RUN-ECDLP-07aad1; 500,000 replicates within four field-bit strata. Each control pools two prime arms.

All 150,864 jobs completed; main rho censors = 0. All recovered scalars and collision equations passed arithmetic checks. All 768 sham, replay and transported-isomorphism checks and four BSGS easy positive controls passed. 512 actual transported endpoint jobs were also checked; full identities accompany the report.

Rho endpoint: search group calls/sqrt(ell), capped at normalized horizon 20; setup and verification remain separate. Each comparison pools two control arms per slot, then averages eight slot log-ratios per block. BSGS uses setup+search normalized by M+ell/(2M), M≤4096, with table memory retained.

Every one of 11,536 BSGS jobs matches exact count identities: setup=M+popcount(M)+bit_length(M)−1; search=floor(q/M). Its empirical contrast includes target position and finite ell/table-size effects; it does not establish extra resistance from field-prime form.

Implications and scope

The unconditional proxy (0.5 log2 largest prime factor of curve order) differs by 0.0256 bits, S minus pooled random controls. Its marginal 99.9% interval is −0.1061–0.1555. This is not a certified security bound.

The arithmetic panel has 6,480 exact-output-equal observations. Python folding and builtin modular arithmetic are implementation comparisons. Inversion uses identical builtin pow in both paths. report.md and analysis/summary.json retain every timing ratio and higher-confidence interval; production C/assembly arithmetic is untested.

Generic rho collision work depends mainly on subgroup size. Sparse binary expressions can affect field reduction. Equal-order cyclic groups being abstractly isomorphic supplies no efficient coordinate map between arbitrary curves. The implemented opaque interface limits this solver’s coordinate access; it is not a formally enforced oracle against an adversarial program.

Scope is the declared finite sparse-prime catalog, uniform nonsingular coefficient models, matching rules and implemented toy attacks. This does not settle deployed ECDLP, structural attacks, all Solinas primes, number-field rank or unrelated prime families. No scientific claim, sealed bound or canonical frontier is promoted.

Artifacts and provenance

See manifest.json and contract.json for preselection source commit, hashes, seeds and exact commands; execution-result.json for actual phase outcomes; analysis/summary.json for every estimate; report/all-curve-identities.csv for 3,260 full EC1 identities; report.md for links and detailed limitations. All raw outputs are retained. Committed per-block matching shards reconstruct the exact retained large array; this changes storage only. No independent scientific review was obtained.